Exercises

Review Exercises

Decimals

Name Decimals

In the following exercises, name each decimal.

534. 0.8

535. 0.375

536. 0.007

537. 5.24

538. −12.5632

539. −4.09

Write Decimals

In the following exercises, write as a decimal.

540. three tenths

541. nine hundredths

542. twenty-seven hundredths

543. ten and thirty-five thousandths

544. negative twenty and three tenths

545. negative five hundredths

Convert Decimals to Fractions or Mixed Numbers

In the following exercises, convert each decimal to a fraction. Simplify the answer if possible.

546. 0.43

547. 0.825

548. 9.7

549. 3.64

Locate Decimals on the Number Line

550. ⓐ 0.6
ⓑ −0.9
ⓒ 2.2
ⓓ −1.3

Order Decimals

In the following exercises, order each of the following pairs of numbers, using $<$ or $>$.

551. 0.6 ___ 0.8

552. 0.2 ___ 0.15

553. 0.803 ____ 0.83

554. −0.56 ____ −0.562

Round Decimals

In the following exercises, round each number to the nearest: ⓐ hundredth ⓑ tenth ⓒ whole number.

555. 12.529

556. 4.8447

557. 5.897

Decimal Operations

Add and Subtract Decimals

In the following exercises, add or subtract.

558. $5.75 + 8.46$

559. $32.89 - 8.22$

560. $24 - 19.31$

561. $10.2 + 14.631$

562. $-6.4 + (-2.9)$

563. $1.83 - 4.2$

Multiply Decimals

In the following exercises, multiply.

564. $(0.3)(0.7)$

565. $(-6.4)(0.25)$

566. $(-3.35)(-12.7)$

567. $(15.4)(1000)$

Divide Decimals

In the following exercises, divide.

568. $0.48 \div 6$

569. $4.32 \div 24$

570. $\$6.29 \div 12$

571. $(-0.8) \div (-0.2)$

572. $1.65 \div 0.15$

573. $9 \div 0.045$

Use Decimals in Money Applications

In the following exercises, use the strategy for applications to solve.

574. Miranda got $\$40$ from her ATM. She spent $\$9.32$ on lunch and $\$16.99$ on a book. How much money did she have left? Round to the nearest cent if necessary.

575. Jessie put 8 gallons of gas in her car. One gallon of gas costs $\$3.528$. How much did Jessie owe for all the gas?

576. A pack of 16 water bottles cost $\$6.72$. How much did each bottle cost?

577. Alice bought a roll of paper towels that cost $\$2.49$. She had a coupon for $\$0.35$ off, and the store doubled the coupon. How much did Alice pay for the paper towels?

Decimals and Fractions

Convert Fractions to Decimals

In the following exercises, convert each fraction to a decimal.

578. $\frac{3}{5}$

579. $\frac{7}{8}$

580. $-\frac{19}{20}$

581. $-\frac{21}{4}$

582. $\frac{1}{3}$

583. $\frac{6}{11}$

Order Decimals and Fractions

In the following exercises, order each pair of numbers, using $<$ or $>$.

584. $\frac{1}{2}$ ___ 0.2

585. $\frac{3}{5}$ ___ $0.$

586. $-\frac{7}{8}$ ___ -0.84

587. $-\frac{5}{12}$ ___ -0.42

588. 0.625 ___ $\frac{13}{20}$

589. 0.33 ___ $\frac{5}{16}$

In the following exercises, write each set of numbers in order from least to greatest.

590. $\frac{2}{3}, \frac{17}{20}, 0.65$

591. $\frac{7}{9}, 0.75, \frac{11}{15}$

Simplify Expressions Using the Order of Operations

In the following exercises, simplify

592. $4(10.3 - 5.8)$

593. $\frac{3}{4}(15.44 - 7.4)$

594. $30 \div (0.45 + 0.15)$

595. $1.6 + \frac{3}{8}$

596. $52(0.5) + (0.4)^2$

597. $-\frac{2}{5} \cdot \frac{9}{10} + 0.14$

Find the Circumference and Area of Circles

In the following exercises, approximate the ⓐ circumference and ⓑ area of each circle.

598. radius $= 6$ in.

599. radius $= 3.5$ ft.

600. radius $= \frac{7}{33}$ m

601. diameter $= 11$ cm

Solve Equations with Decimals

Determine Whether a Decimal is a Solution of an Equation

In the following exercises, determine whether the each number is a solution of the given equation.

602. $x - 0.4 = 2.1$
ⓐ $x = 1.7$ ⓑ $x = 2.5$

603. $y + 3.2 = -1.5$
ⓐ $y = 1.7$ ⓑ $y = -4.7$

604. $\frac{u}{2.5} = -12.5$
ⓐ $u = -5$ ⓑ $u = -31.25$

605. $0.45v = -40.5$
ⓐ $v = -18.225$ ⓑ $v = -90$

Solve Equations with Decimals

In the following exercises, solve.

606. $m + 3.8 = 7.5$

607. $h + 5.91 = 2.4$

608. $a + 2.26 = -1.1$

609. $p - 4.3 = -1.65$

610. $x - 0.24 = -8.6$

611. $j - 7.42 = -3.7$

612. $0.6p = 13.2$

613. $-8.6x = 34.4$

614. $-22.32 = -2.4z$

615. $\frac{a}{0.3} = -24$

616. $\frac{p}{-7} = -4.2$

617. $\frac{s}{-2.5} = -10$

Translate to an Equation and Solve

In the following exercises, translate and solve.

618. The difference of n and 15.2 is 4.4.

619. The product of -5.9 and x is -3.54.

620. The quotient of y and -1.8 is -9.

621. The sum of m and (-4.03) is 6.8.

Averages and Probability

Find the Mean of a Set of Numbers

In the following exercises, find the mean of the numbers.

622. $2, 4, 1, 0, 1,$ and 1

623. $270, $310.50, $243.75, and $252.15

624. Each workday last week, Yoshie kept track of the number of minutes she had to wait for the bus. She waited $3, 0, 8, 1,$ and 8 minutes. Find the mean.

625. In the last three months, Raul's water bills were $31.45, $48.76, and $42.60. Find the mean.

Find the Median of a Set of Numbers

In the following exercises, find the median.

626. 41, 45, 32, 60, 58

627. 25, 23, 24, 26, 29, 19, 18, 32

628. The ages of the eight men in Jerry's model train club are 52, 63, 45, 51, 55, 75, 60, and 59. Find the median age.

629. The number of clients at Miranda's beauty salon each weekday last week were 18, 7, 12, 16, and 20. Find the median number of clients.

Find the Mode of a Set of Numbers

In the following exercises, identify the mode of the numbers.

630. 6, 4, 4, 5, 6, 6, 4, 4, 4, 3, 5

631. The number of siblings of a group of students: 2, 0, 3, 2, 4, 1, 6, 5, 4, 1, 2, 3

Use the Basic Definition of Probability

In the following exercises, solve. (Round decimals to three places.)

632. The Sustainability Club sells 200 tickets to a raffle, and Albert buys one ticket. One ticket will be selected at random to win the grand prize. Find the probability Albert will win the grand prize. Express your answer as a fraction and as a decimal.

633. Luc has to read 3 novels and 12 short stories for his literature class. The professor will choose one reading at random for the final exam. Find the probability that the professor will choose a novel for the final exam. Express your answer as a fraction and as a decimal.

Ratios and Rate

Write a Ratio as a Fraction

In the following exercises, write each ratio as a fraction. Simplify the answer if possible.

634. 28 to 40

635. 56 to 32

636. 3.5 to 0.5

637. 1.2 to 1.8

638. $1\frac{3}{4}$ to $1\frac{5}{8}$

639. $2\frac{1}{3}$ to $5\frac{1}{4}$

640. 64 ounces to 30 ounces

641. 28 inches to 3 feet

Write a Rate as a Fraction

In the following exercises, write each rate as a fraction. Simplify the answer if possible.

642. 180 calories per 8 ounces

643. 90 pounds per 7.5 square inches

644. 126 miles in 4 hours

645. $612.50 for 35 hours

Find Unit Rates

In the following exercises, find the unit rate.

646. 180 calories per 8 ounces

647. 90 pounds per 7.5 square inches

648. 126 miles in 4 hours

649. $612.50 for 35 hours

Find Unit Price

In the following exercises, find the unit price.

650. t-shirts: 3 for $8.97

651. Highlighters: 6 for $2.52

652. An office supply store sells a box of pens for $11. The box contains 12 pens. How much does each pen cost?

653. Anna bought a pack of 8 kitchen towels for $13.20. How much did each towel cost? Round to the nearest cent if necessary.

In the following exercises, find each unit price and then determine the better buy.

654. Shampoo: 12 ounces for $4.29 or 22 ounces for $7.29?

655. Vitamins: 60 tablets for $6.49 or 100 for $11.99?

Translate Phrases to Expressions with Fractions

In the following exercises, translate the English phrase into an algebraic expression.

656. 535 miles per h hours

657. a adults to 45 children

658. the ratio of $4y$ and the difference of x and 10

659. the ratio of 19 and the sum of 3 and n

Simplify and Use Square Roots

Simplify Expressions with Square Roots

In the following exercises, simplify.

660. $\sqrt{64}$

661. $\sqrt{144}$

662. $-\sqrt{25}$

663. $-\sqrt{81}$

664. $\sqrt{-9}$

665. $\sqrt{-36}$

666. $\sqrt{64}+\sqrt{225}$

667. $\sqrt{64+225}$

Estimate Square Roots

In the following exercises, estimate each square root between two consecutive whole numbers.

668. $\sqrt{28}$

669. $\sqrt{155}$

Approximate Square Roots

In the following exercises, approximate each square root and round to two decimal places.

670. $\sqrt{15}$

671. $\sqrt{57}$

Simplify Variable Expressions with Square Roots

In the following exercises, simplify. (Assume all variables are greater than or equal to zero.)

672. $\sqrt{q^2}$

673. $\sqrt{64b^2}$

674. $-\sqrt{121a^2}$

675. $\sqrt{225m^2n^2}$

676. $-\sqrt{100q^2}$

677. $\sqrt{49y^2}$

678. $\sqrt{4a^2b^2}$

679. $\sqrt{121c^2d^2}$

Use Square Roots in Applications

In the following exercises, solve. Round to one decimal place.

680. **Art** Diego has 225 square inch tiles. He wants to use them to make a square mosaic. How long can each side of the mosaic be?

681. **Landscaping** Janet wants to plant a square flower garden in her yard. She has enough topsoil to cover an area of 30 square feet. How long can a side of the flower garden be?

682. **Gravity** A hiker dropped a granola bar from a lookout spot 576 feet above a valley. How long did it take the granola bar to reach the valley floor?

683. **Accident investigation** The skid marks of a car involved in an accident were 216 feet. How fast had the car been going before applying the brakes?

Practice Test

684. Write six and thirty-four thousandths as a decimal.

685. Write 1.73 as a fraction.

686. Write $\frac{5}{8}$ as a decimal.

687. Round 16.749 to the nearest (a) tenth (b) hundredth (c) whole number

688. Write the numbers $\frac{4}{5}, -0.1, 0.804, \frac{2}{9}, -7.4, 0.21$ in order from smallest to largest.

In the following exercises, simplify each expression.

689. $15.4 + 3.02$

690. $20 - 5.71$

691. $(0.64)(0.3)$

692. $(-4.2)(100)$

693. $0.96 \div (-12)$

694. $-5 \div 0.025$

695. $-0.6 \div (-0.3)$

696. $(0.7)^2$

697. $24 \div (0.1 + 0.02)$

698. $4(10.3 - 5.8)$

699. $1.6 + \frac{3}{8}$

700. $\frac{2}{3}(14.65 - 4.6)$

In the following exercises, solve.

701. $m + 3.7 = 2.5$

702. $\frac{h}{0.5} = 4.38$

703. $-6.5y = -57.2$

704. $1.94 = a - 2.6$

705. Three friends went out to dinner and agreed to split the bill evenly. The bill was $79.35. How much should each person pay?

706. A circle has radius 12. Find the (a) circumference and (b) area. [Use 3.14 for π.]

707. The ages, in months, of 10 children in a preschool class are:
$55, 55, 50, 51, 52, 50, 53, 51, 55, 49$
Find the (a) mean (b) median (c) mode

708. Of the 16 nurses in Doreen's department, 12 are women and 4 are men. One of the nurses will be assigned at random to work an extra shift next week. (a) Find the probability a woman nurse will be assigned the extra shift. (b) Convert the fraction to a decimal.

709. Find each unit price and then the better buy.
Laundry detergent: 64 ounces for $\$10.99$ or 48 ounces for $\$8.49$

In the following exercises, simplify.

710. $\sqrt{36+64}$

711. $\sqrt{144n^2}$

712. Estimate $\sqrt{54}$ to between two whole numbers.

713. Yanet wants a square patio in her backyard. She has 225 square feet of tile. How long can a side of the patio be?